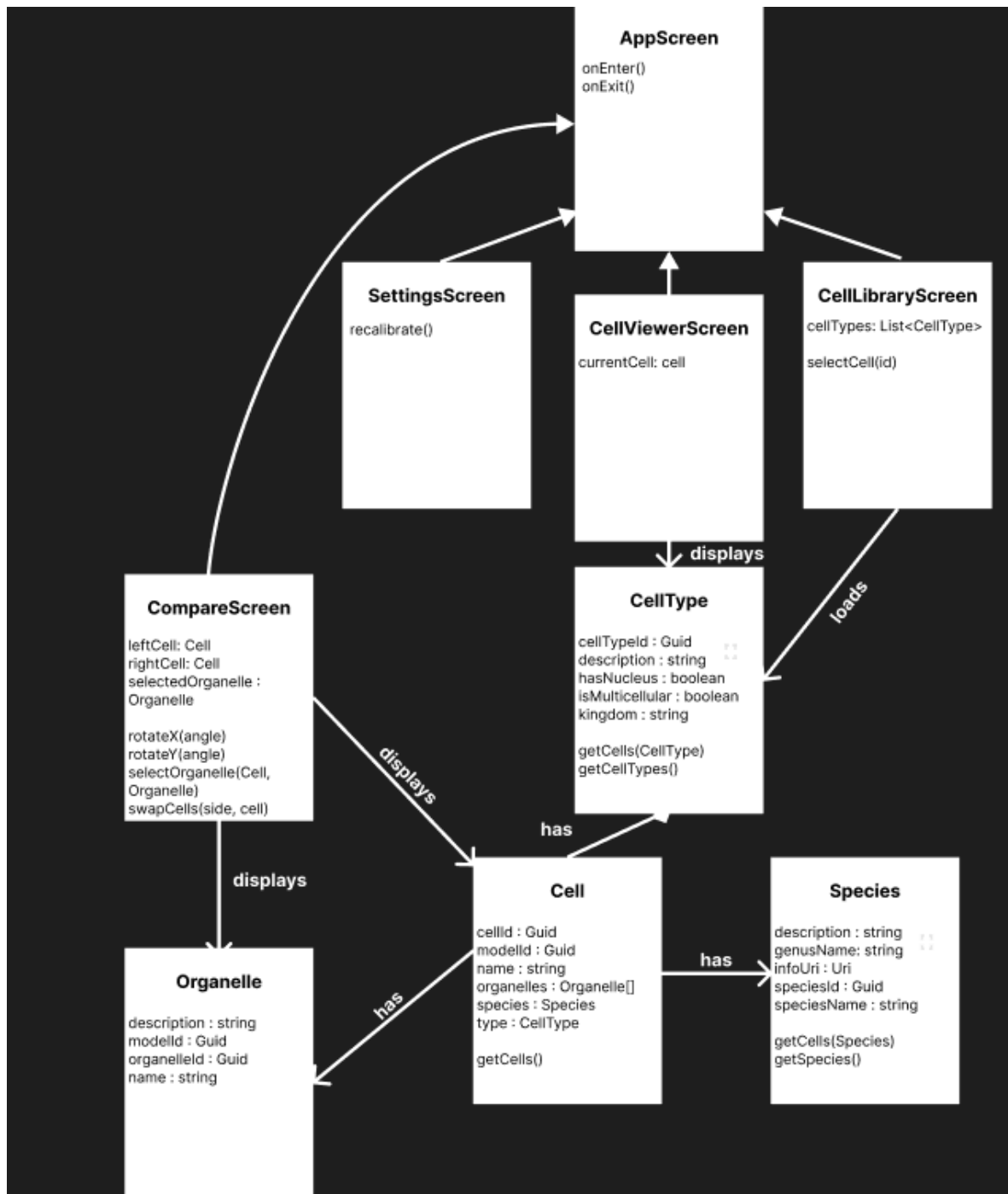


Team: From Null

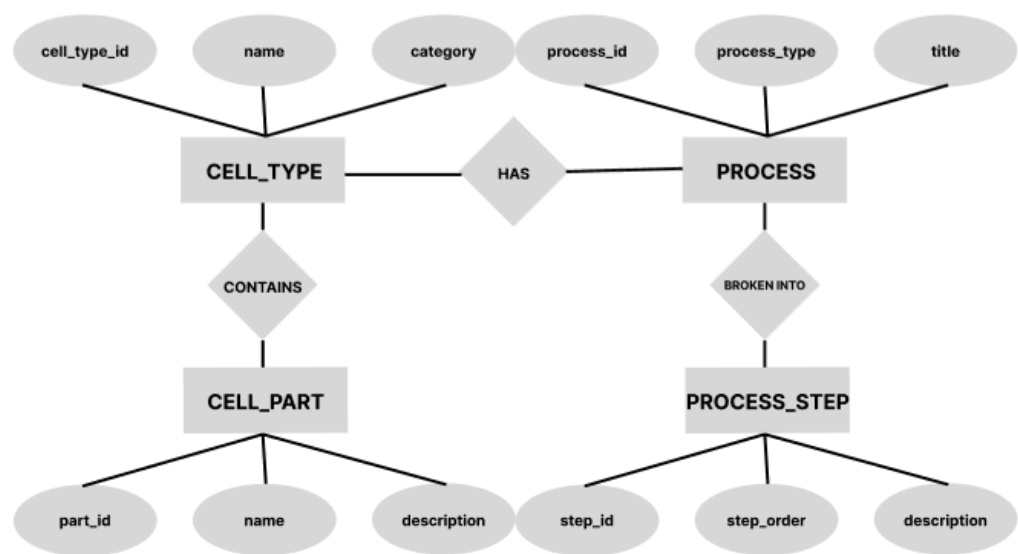
Gerardo Martinez, Michael Nelson, Nolan Wilkins, John Shaffer

<i>Revision Number</i>	<i>Revision Date</i>	<i>Summary of Changes</i>	<i>Author(s)</i>
0.2	9/25/2026	Replaced some UI wireframes with early development screenshots.	John Shaffer
0.1	09/22/2026	Initial Design Document	Gerardo Martinez, Michael Nelson, Nolan Wilkins, John Shaffer

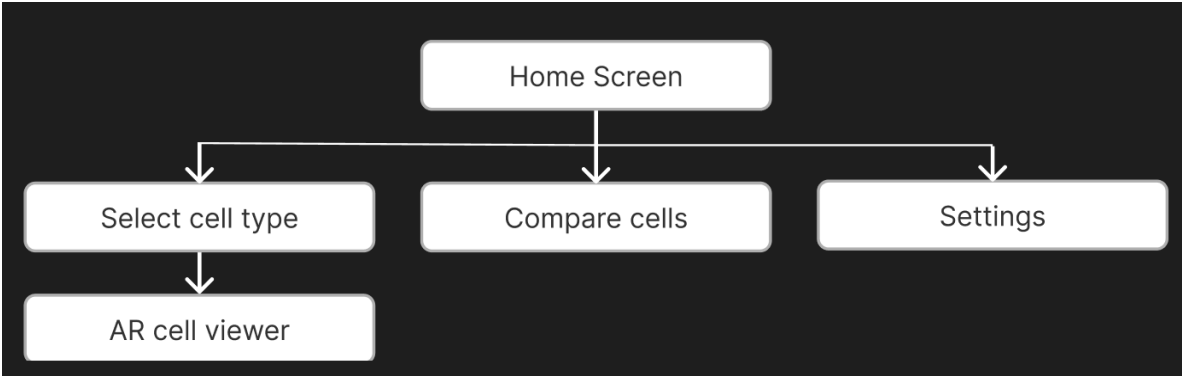
Class Diagram(s)



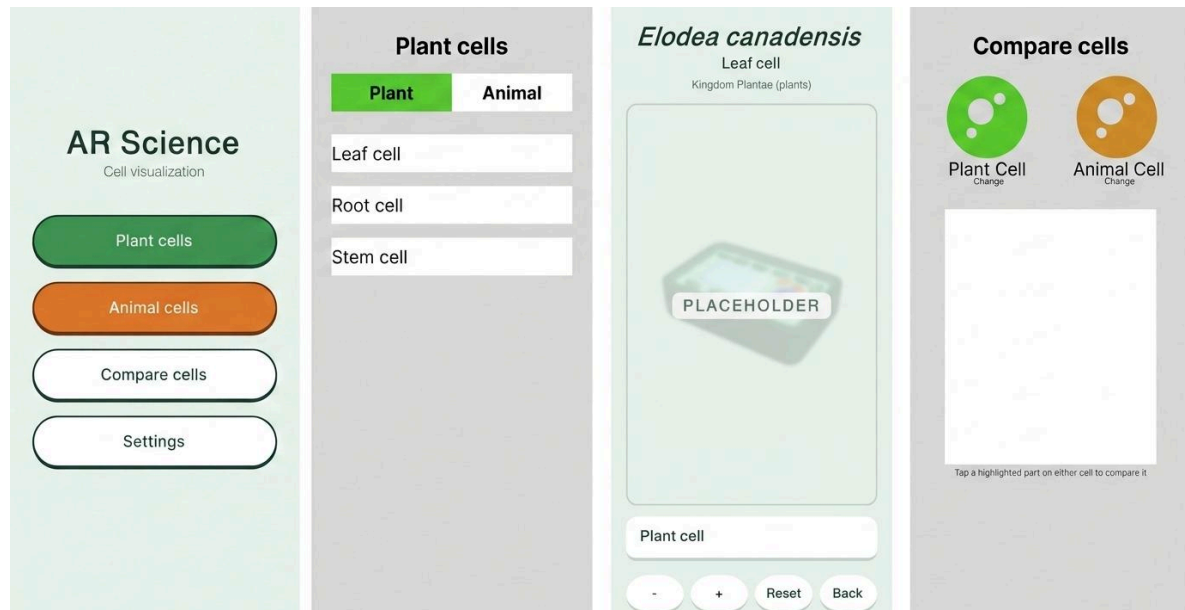
ER Diagram(s)



Information Architecture Diagram



User Interface Wireframe(s)/Screenshot(s)



Tapping a category opens the list type.

Tapping compare cells opens the compare cells screen.

Tapping Animal, or Plant if you are in the Animal screen, brings you to the respective cell type.

Clicking one of the cell options lets you view that cell.

The - button lets you place the model.

The + button lets you zoom in and out, also available through the touchscreen.

The next button lets you turn labels on and off.

The Re button lets you recalibrate.

The back arrow takes you back.

Design Summary

The AR Science Application is a mobile app for both iOS and Android designed to help both upper elementary and middle school students learn various things about cell biology through AR.

UI and Navigation: First there is a homescreen that leads to three main paths, selecting cell types, comparing cells, or managing the app settings (for now just calibration)

AppScreen serves as the main screen where you can get to any page.

CellLibraryScreen allows users to pick specific plant or animals cells

CellViewerScreen renders the selected model in AR

CompareScreen allows the comparison of two cells.

SettingsScreen enables AR recalibration (probably going to have actual settings later)

Cell is our main variable type with cellId, modelId, name, type, species and holds a collection of organelles: Organelles[]

CellType is for the different types of cells, it has cellTypeId, description, hasNucleus, isMulticellular, kingdom and also has getCells(CellType) and getCellTypes()

Species holds information about the species with description, genusName, infoUri, speciesId, speciesName and also has getCells(Species) and get(Species)

CELL_TYPE CONTAINS CELL_PART, which maps internal organelle and their descriptions.

CELL_TYPE HAS PROCESS, which maps the cellular mechanisms like energy flow or cell division which is BROKEN INTO PROCESS_STEP records.

Design Rationale

We are planning on developing this using Unity AR Foundation since it supports AR development and works for both iOS and Android devices.

Developing in their native development, like Swift for iOS would offer a little bit of performance gain but doing it again for Android isn't worth the very long time commitment.

We are planning on using a hybrid of both touch screen and AR for the UI.

We are doing this to make it so you don't always have to manually go around the object with your phone and can for example rotate it without physically moving.

For comparison mode for now we are only displaying one at a time.

We are doing this because of performance improvements. The newer phones can probably handle two objects at once but two might be an issue if we decide to port to older phones too. We will see based on our tests.